

C926

Earth: Inside and Out

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A research ship on its 12th expedition, crewed by 48 scientists, drills seafloor samples along a line from the Mid-Atlantic Ridge toward North America. What pattern in rock age should the team find?

- A. Youngest at the ridge, getting older toward the continent
- B. Oldest at the ridge, getting younger toward the continent
- C. Older than the oldest continental rocks
- D. About the same age everywhere

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Continental rocks older than 4 billion years exist, yet no seafloor is much older than about 180 million years. Why?

- A. Old ocean crust is subducted and recycled into the mantle.
- B. Ocean currents erode old seafloor away.
- C. Continents are recycled faster than ocean floor.
- D. All ocean floor formed at once about 180 million years ago.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A survey plane from a coastal-mapping agency tows a magnetometer across a mid-ocean ridge at 300 m altitude and finds bands of normal and reversed polarity that mirror each other on both sides. What best explains the symmetry?

- A. Changes in ocean temperature create the bands.
- B. New crust records the field as it forms at the ridge, then moves away on both sides.
- C. Lava flows from the continents toward the ridge.
- D. Earth's field has never reversed; the bands just reflect rock type.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Basalt 60 km from a ridge axis is dated at 2 million years old by a lab that charges \$900 per sample. At what average rate has this side of the plate moved away from the ridge?

- A. 120 km per million years
- B. 3 cm/yr
- C. 6 cm/yr
- D. 0.3 cm/yr

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A plate moves away from a ridge at 5 cm/yr. How far from the ridge is crust that is 4 million years old?

- A. 0.8 km
- B. 400 km
- C. 20 km
- D. 200 km

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

In a volcanic island chain that draws 8 million tourists a year, the only active volcano is at the southeast end, and islands get steadily older toward the northwest. The hot spot is fixed. Which way is the plate moving?

- A. Northwest
- B. It cannot be told, since chains always form all at once.
- C. Southeast
- D. The hot spot is moving northwest; the plate is still.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

About 10,000 years after a 2-km-thick ice sheet melted, land around a bay that now supports a fishing port with three canneries is still rising about 1 cm a year. Why?

- A. Solid land cannot move; the measurements must be wrong.
- B. New magma is pushing the land up.
- C. Isostatic rebound: the crust is slowly rising after the ice's weight was removed.
- D. Sea level is falling, so the land only appears to rise.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A geophysics textbook with a glossy cover describes the asthenosphere, below the lithosphere, where S-waves slow down but still pass through. Which statement is correct?

- A. The lithosphere is the crust alone.
- B. The asthenosphere is part of the outer core.
- C. The asthenosphere is solid but weak and flows slowly; the lithosphere is the crust plus the uppermost mantle.
- D. The asthenosphere is a layer of liquid magma.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which layer of Earth is liquid?

- A. The inner core
- B. The entire mantle
- C. The asthenosphere
- D. The outer core

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Shells of tiny marine organisms in a sediment core, stored in a repository that holds 20 km of cores, show $\delta^{18}\text{O}$ rising sharply at one level. What does this most likely indicate?

- A. A burst of volcanic activity
- B. A colder climate with larger ice sheets
- C. Nothing about climate; isotopes never reflect climate.
- D. A warmer climate with less ice

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

During a warm interglacial, ice sheets shrink. What happens to the $^{18}\text{O}/^{16}\text{O}$ ratio of seawater?

- A. It increases.
- B. ^{18}O disappears from seawater.
- C. It decreases.
- D. It stays the same.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A poster sold for \$15 at a museum gift shop compresses Earth’s 4.6-billion-year history into one 24-hour day. About how much of that day would the roughly 300,000 years of Homo sapiens fill?

- A.** About 1 hour
- B.** About 15 seconds
- C.** About 6 minutes
- D.** About 6 seconds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A research ship ~~on its 12th expedition, crewed by 48 scientists,~~ drills seafloor samples along a line from the Mid-Atlantic Ridge toward North America. What pattern in rock age should the team find?

Target word: age pattern **Decides:** New crust forms at the ridge.

A	Correct	Crust moves away from where it formed.
B	Inverted operation	Reverses where crust is made.
C	Confusion set	Seafloor is far younger; it is recycled by subduction.
D	Dropped element	Drops seafloor spreading.

2. Answer A Concept 1 • transfer

Continental rocks older than 4 billion years exist, yet no seafloor is much older than about 180 million years. Why?

Target word: why seafloor is young **Decides:** Oceanic lithosphere sinks at trenches.

A	Correct	Dense oceanic plates sink; buoyant continents stay.
B	Confusion set	Deep currents cannot remove crust.
C	Inverted operation	Continents resist subduction; that is why they are older.
D	Adjacent category	Seafloor forms continuously at ridges.

3. Answer B Concept 2

A survey plane ~~from a coastal mapping agency~~ tows a magnetometer across a mid-ocean ridge ~~at 300 m altitude~~ and finds bands of normal and reversed polarity that mirror each other on both sides. What best explains the symmetry?

Target word: cause of mirror stripes **Decides:** Crust records the field at the ridge and splits both ways.

A	Adjacent category	The stripes are recorded in the rock's magnetism.
B	Correct	Each reversal is carried out symmetrically.
C	Inverted operation	Crust moves away from the ridge, not toward it.
D	Absolute language	Reversals are well documented on land and sea.

4. Answer B Concept 3

Basalt 60 km from a ridge axis is dated at 2 million years old ~~by a lab that charges \$900 per sample~~. At what average rate has this side of the plate moved away from the ridge?

Target word: one-side rate **Decides:** $60 \text{ km} \div 2 \text{ million years} = 30 \text{ km per million years}$.

A	Inverted operation	Multiplies distance by age.
B	Correct	$6,000,000 \text{ cm} \div 2,000,000 \text{ yr}$.
C	Confusion set	Doubles to the full spreading rate, which was not asked.
D	Dropped element	Drops a factor of 10 in the km-to-cm conversion.

5. Answer D Concept 3 • transfer

A plate moves away from a ridge at 5 cm/yr. How far from the ridge is crust that is 4 million years old?

Target word: solve for distance **Decides:** Distance = rate × time.

A	Inverted operation	Divides instead of multiplying.
B	Confusion set	Doubles for both sides of the ridge.
C	Dropped element	Drops a factor of 10 in the conversion.
D	Correct	$5 \times 4,000,000 = 20,000,000 \text{ cm} = 200 \text{ km}.$

6. Answer A Concept 4

In a volcanic island chain ~~that draws 8 million tourists a year~~, the only active volcano is at the southeast end, and islands get steadily older toward the northwest. The hot spot is fixed. Which way is the plate moving?

Target word: plate direction **Decides:** Older volcanoes are carried in the direction of motion.

A	Correct	The plate carries older islands away from the hot spot toward the northwest.
B	Absolute language	The age sequence shows gradual formation.
C	Inverted operation	Points toward the youngest volcano.
D	Confusion set	The stem says the hot spot is fixed.

7. Answer C Concept 5

About 10,000 years after a 2-km-thick ice sheet melted, land around a bay ~~that now supports a fishing port with three canneries~~ is still rising about 1 cm a year. Why?

Target word: cause of uplift **Decides:** The ice weight is gone.

A	Absolute language	Crust moves vertically in response to loads.
B	Adjacent category	No volcanic source is described.
C	Correct	The mantle flows back slowly beneath it.
D	Confusion set	Confuses global sea-level change with vertical land motion.

8. Answer C Concept 6

A geophysics textbook ~~with a glossy cover~~ describes the asthenosphere, below the lithosphere, where S-waves slow down but still pass through. Which statement is correct?

Target word: state of the asthenosphere **Decides:** S-waves pass through, so it is solid.

A	Dropped element	Drops the rigid uppermost mantle.
B	Adjacent category	It lies in the upper mantle.
C	Correct	S-waves do not travel through liquids.
D	Confusion set	S-waves would stop in a liquid.

9. Answer D Concept 6 • transfer

Which layer of Earth is liquid?

Target word: the liquid layer **Decides:** S-waves do not pass the outer core.

A	Inverted operation	Extreme pressure keeps it solid.
B	Absolute language	The mantle is almost entirely solid rock.
C	Confusion set	Weak and plastic, but solid.
D	Correct	The S-wave shadow zone shows it is liquid.

10. Answer B Concept 7

Shells of tiny marine organisms in a sediment core, ~~stored in a repository that holds 20 km of cores,~~ show $\delta^{18}\text{O}$ rising sharply at one level. What does this most likely indicate?

Target word: climate meaning **Decides:** Ice stores ^{16}O , leaving seawater richer in ^{18}O .

A	Adjacent category	Oxygen isotopes in shells track ice volume and temperature.
B	Correct	More ice leaves oceans and shells enriched in ^{18}O .
C	Absolute language	They are a standard climate proxy.
D	Inverted operation	Warmth releases ^{16}O back to the sea and lowers $\delta^{18}\text{O}$.

11. Answer C Concept 7 • transfer

During a warm interglacial, ice sheets shrink. What happens to the $^{18}\text{O}/^{16}\text{O}$ ratio of seawater?

Target word: direction of change **Decides:** Meltwater rich in ^{16}O returns to the ocean.

A	Inverted operation	That happens when ice grows.
B	Absolute language	Ratios shift slightly; neither isotope vanishes.
C	Correct	Added ^{16}O dilutes the ^{18}O .
D	Dropped element	Drops the meltwater's isotope signature.

12. Answer D Concept 8

A poster ~~sold for \$15 at a museum gift shop~~ compresses Earth's 4.6-billion-year history into one 24-hour day. About how much of that day would the roughly 300,000 years of Homo sapiens fill?

Target word: scaled time **Decides:** $300,000 \div 4.6 \text{ billion} \times 86,400 \text{ s}$.

A	Confusion set	The common overestimate of human history.
B	Camouflage	Borrows the poster's price.
C	Dropped element	Drops the conversion from minutes to seconds (off by 60).
D	Correct	$\approx 5.6 \text{ s}$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Seafloor age pattern	Ocean crust is youngest at mid-ocean ridges and older with distance; no seafloor is much older than about 180-200 million years because it is subducted.	Thinking the ocean floor is the oldest crust.	Samples collected along a line away from a ridge.
2	Magnetic striping	New crust records Earth's field as it forms at the ridge and splits apart, producing mirror-image stripes of normal and reversed polarity.	Explaining stripes by rock type or temperature.	Symmetric magnetic anomalies on both sides of a ridge.
3	Spreading rate	Rate = distance from the ridge ÷ age of the crust (one side); the full spreading rate is twice that.	Unit slips (km per million years vs. cm per year) or doubling when one side is asked.	A distance from the ridge and an age.
4	Hot-spot tracks	A fixed hot spot under a moving plate leaves a volcano chain that gets older in the direction the plate moves.	Reversing the direction, or thinking the hot spot moves.	An island chain with an active volcano at one end.
5	Isostasy	Crust floats on the mantle; adding weight (ice) pushes it down and removing weight lets it rebound slowly.	Crediting new magma or falling sea level for rising land.	Land still rising thousands of years after ice melted.
6	Mechanical vs. compositional layers	The lithosphere is the rigid crust plus uppermost mantle; the asthenosphere below is solid but weak and flows slowly; only the outer core is liquid.	Picturing the mantle or asthenosphere as molten.	"Which layer is liquid?" or S-waves passing through.
7	Oxygen-isotope climate records	Ice sheets lock up light ^{16}O , so during cold, icy times seawater and shells are richer in ^{18}O (higher $\delta^{18}\text{O}$).	Reading higher $\delta^{18}\text{O}$ as warmer.	$\delta^{18}\text{O}$ in shells or ice cores.
8	Deep time	Earth is about 4.6 billion years old; human history is a vanishingly small fraction of that span.	Overestimating how much of Earth history humans occupy.	Earth history compressed into a day or a year.

Course scope (public description): Earth: Inside and Out explores the ways in which our dynamic planet evolved, and the processes and systems that continue to shape it. Though the geologic record is incredibly ancient, it has only been studied intensely since the end of the 19th century. Since then, research in fields such as geologic time, plate tectonics, climate change, exploration of the deep-sea floor, and the inner earth have vastly increased our understanding of geological processes.